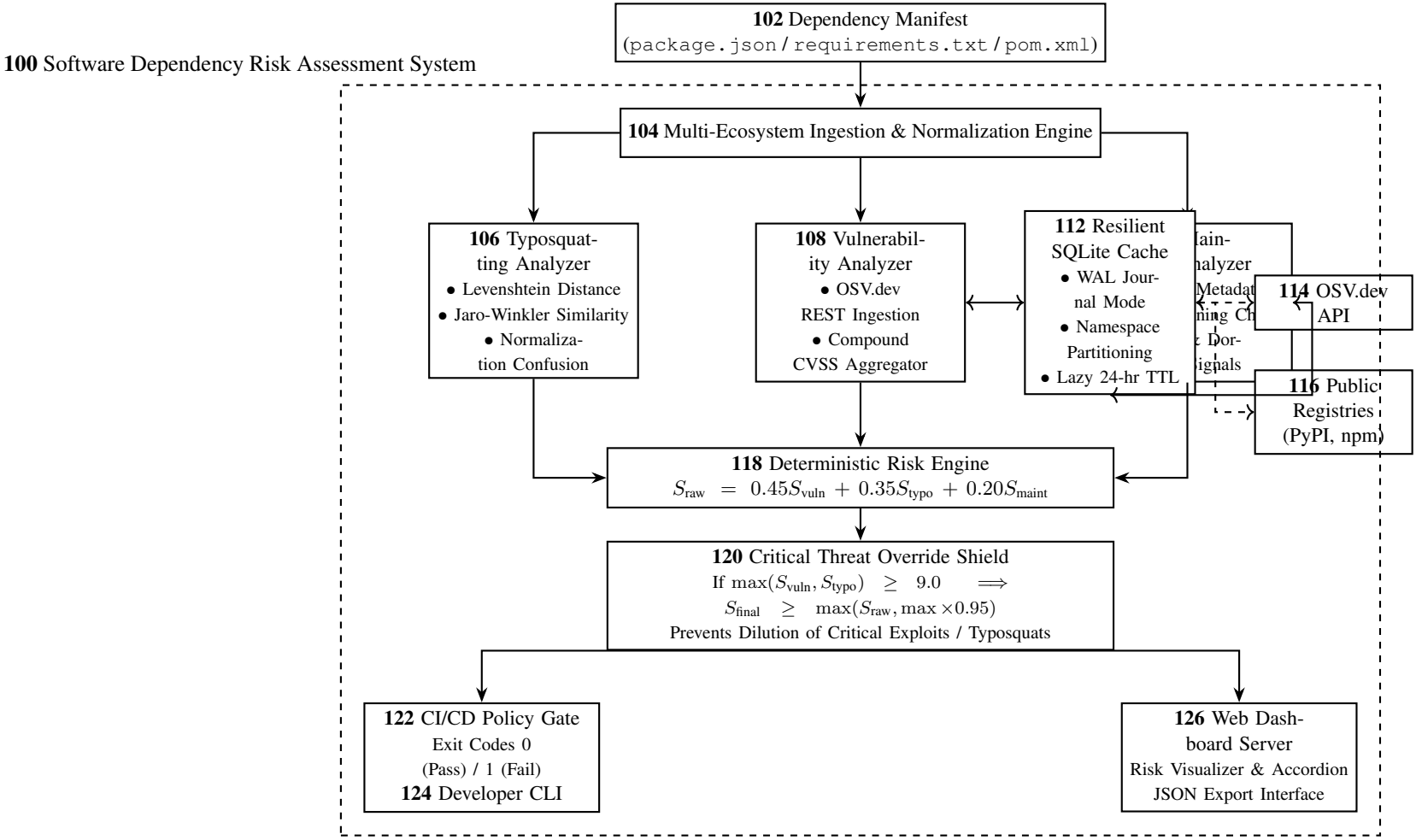

100 Software Dependency Risk Assessment System
102 Dependency Manifest
(package.json / requirements.txt / pom.xml)
104 Multi-Ecosystem Ingestion & Normalization Engine
106 Typosquat-
ting Analyzer
• Levenshtein Distance
• Jaro-Winkler Similarity
• Normaliza-
tion Confusion
108 Vulnerabil-
ity Analyzer
• OSV.dev
REST Ingestion
• Compound
CVSS Aggregator
112 Resilient
SQLite Cache
• WAL Jour-
nal Mode
• Namespace
Partitioning
• Lazy 24-hr TTL
114 OSV.dev
API
116 Public
Registries
(PyPI, npm)
118 Deterministic Risk Engine
$S_{\text{raw}} = 0.45S_{\text{vuln}} + 0.35S_{\text{typo}} + 0.20S_{\text{maint}}$
120 Critical Threat Override Shield
If $\max(S_{\text{vuln}}, S_{\text{typo}}) \geq 9.0 \implies$
$S_{\text{final}} \geq \max(S_{\text{raw}}, \max \times 0.95)$
Prevents Dilution of Critical Exploits / Typosquats
122 CI/CD Policy Gate
Exit Codes 0
(Pass) / 1 (Fail)
124 Developer CLI
126 Web Dash-
board Server
Risk Visualizer & Accordion
JSON Export Interface